F24-047-D-ScholarChain
AI-Driven Student Loaning Platform on Blockchain

Project Team

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Chapter 1

Introduction

The following project proposal focuses on ScholarChain, which is an AI-driven Student Loan System based on the Blockchain that aims to provide interest-free loans to students to help them pursue their educational ambitions. ScholarChain revolves around revolving crowd-sourced funds via social welfare and charitable organizations, allowing such organizations to have a transparent and secure platform to connect with those who need their services. The proposal aims to provide a detailed description of the system and its core features, as well as the motivation behind the project.

1.1 Problem Statement

Quality education is a fundamental human right. However, this human right is slipping out of the reach of a lot of people. This is due to the extremely high tuition fee that is not affordable by most students. This continuous rise in costs is causing students to give up on their academic aspirations entirely, either by dropping out halfway or choosing not to pursue a higher education at all. While financial support systems do exist, they are inadequate and often come with their own unique set of problems that can become a burden on the students in their own way. Loans are accompanied by strict repayment schedules and high interest rates which can lead to long-term debts. According to [1], the application process for such financial aid services is also very complex which can deter many students and have stringent eligibility criteria that are not very easily met. There is also a major lack of transparency in existing systems which causes distrust amongst the shareholders. Furthermore, charitable organizations face difficulties in effectively distributing funds due to administrative inefficiencies. These issues contribute to turning what should be an effective solution to the escalating costs of quality education into hurdles for the students who need it the most.

1.2 Scope

The purpose of this project is to develop an automated study loaning system that will provide interest-free loans to support students with financially weak backgrounds by connecting them with social welfare organizations through a crowd-sourcing model. Through blockchain technology, the system will guarantee a transparent, secure, and impenetrable loaning process. The system will have its private blockchain to ensure the security of the transactions and transparency only within the system. Furthermore, Blockchain will be used to create chain codes that will automate the entire loan process from agreement to last loan installments and track each transaction. The system will also have a Large Language Model (LLM) which will be used to provide personal assistance to the user during the form-filling process to answer queries. Additionally, LLM will majorly be used in risk assessment of the users' application through analysis of financial documents, and academic records. LLM will also be utilized for personalized plans for loan reimbursement depending on the student's financial conditions. The system will also provide an interface for the donors to donate directly to our system without the involvement of any third parties. Moreover, students will be able to track how much loan has been reimbursed, see transaction logs, and query the chatbot in case of any confusion at any time. Lastly, the system will also automate user reminders about their repayments and automatically update the plans based on the users' timely or late payments.

1.3 Modules

Our project is divided into 3 modules. The modules are listed below.

1.3.1 Module 1 - AI Module

We will be creating a RAG (Retrieval Augmented Generation) which will enhance the user experience by utilizing the power of LLMs along with the context of the documents uploaded by the users. This Module is crucial for the success of the ScholarChain because our system will utilize the analytical power of LLM for the automation of the loaning process This module includes the following features based on AI:

1. Risk Assessment through analysis of financial documents, and academic records and checking if the applicant meets the loan criteria or not.

2. Personal Assistants such as RAGbot or field suggestions/completion to assist the users throughout the form-filling process and reduce errors.

3. Personalized loan repayment plans based on financial conditions to help the student eliminate any financial burden.

4. AI-driven customer support to query questions in case of any confusion to enhance user experience.

1.3.2 Module 2 - Blockchain Module

The blockchain module ensures that the loaning process is automated using chain codes. The implementation of a ledger using the blockchain would provide security and transparency within the system to ensure a tamper-free loaning platform. The following features will be developed under the blockchain module.

1. Immutable Blockchain ledger to log all the transactions to provide a secure and transparent system.

2. chain codes to automate and enforce loan agreements, govern loan transactions, and real-time disbursements and repayment collection.

1.3.3 Module 3 - Web Application

A web application will provide an interface to the students so they can interact with the system, fill out application forms, and track their loan status. It will allow the user for real-time tracking of the loan for the students. An interface for the donors will be made to donate directly into our system using payment gateways. Additionally, the web application will also provide several insights. This module includes the following features:

1. Audit Trails from the donation till loan reimbursement to ensure complete track that the donations are being used for the proposed purpose.

2. Automated financial reports so that students can get insights into the loan status.

3. Automated reminders / Notifications for upcoming or delayed payments to ensure smooth operations.

1.4 User Classes and Characteristics

ScholarChain will be built to be used by a diverse set of users, each of whom will play a crucial role in ensuring the system's effectiveness. This section describes the various user classes that will be interacting with the system and describes their roles and expected interactions within the system.

User class	Description
Students	Students are the primary users for the application as they will be applying for interest-free loans in order to pursue their education. The primary demographic that the system will cater to is students who come from disadvantaged backgrounds and require support to cover their tuition costs. Students will be using the system to apply for loans, provide their documents, and track the application process and repayment schedules. AI assistants will also assist them with any queries that they may have.
Parents	Parents or legal guardians will be acting as co-signers for the students who will be applying for loans. They will be able to use the platform to monitor the repayment schedules as well. They may also co-sign loan agreements alongside their children.
Social Welfare Entities (Donors)	Social Welfare Organizations will serve as the primary financial contributors to the platform. They will provide funding through donations, and the blockchain ledger will ensure the transparency of their contributions. These entities can use the platform to monitor their contributions and see how their funding is being utilized.
System Administrators	The system administrators will be responsible for verifying and approving risk assessment and the repayment plans that will be generated by the AI components of the platform.

Table 1.1: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 High-Level Use Cases

Use Case	Apply for Loan
Actors	Student
Type	Primary
Description	A student will use the ScholarChain platform to submit a loan application which includes filling out personal and academic details and uploading necessary documents. Once the form has been submitted, the student will wait for verification and approval.

Use Case	Provide Funding
Actors	Social Welfare Entity (Donor)
Type	Primary
Description	A donor logs into the ScholarChain platform to provide funding for the revolving fund. They will contribute through secure payment gateways, which will then be recorded on the blockchain for transparency and tracking.

Use Case	Approve Loan
Actors	System Administrator
Type	Primary
Description	The ScholarChain AI module will assess the loan applications by analyzing the documents and details provided by the student. The system administrator will then review the AI assessment and make the final decision regarding the approval of the loan.

Use Case	Track Repayment
Actors	Student, Parent/Guardian
Type	Secondary
Description	Students and parents/guardians will be able to access the ScholarChain platform to monitor loan repayment schedules and view previous payments.

Use Case	Customer Support Interaction
Actors	Student, System Administrator
Type	Primary
Description	Students can access ScholarChain's AI-powered ChatBots to gain assistance. The issue may be escalated to system administrators to provide a resolution in case of a complex issue.

Use Case	Personalized Repayment Plan Generation
Actors	Student
Type	Primary
Description	Once the student has submitted his application and it has been approved, an AI assistant creates a personalized repayment plan for the student based on their financial situation.

Use Case	Automated Loan Disbursement
Actors	System Administrator, Blockchain Ledger
Type	Primary
Description:	Upon approval of a loan, the smart contracts deployed on the blockchain will automatically disburse the designated amount to the student in order to ensure a transparent and tamper-proof process.

Use Case	Verify Loan Agreement
Actors	Student, Parent/Guardian
Type	Secondary
Description	Once the loan has been approved, the agreement will be digitally signed by the student or parent/guardian, which will then be recorded securely on the blockchain ledger.

Use Case	Payment Scheduling and Notifications
Actors	Student
Type	Secondary
Description	The ScholarChain platform sends automated reminders about upcoming payments to the students and provides the option to adjust the repayment plan as needed according to the students current financial condition.

Use Case	Financial Analysis for Risk Assessment
Actors	System Administrator
Typ:	Primary
Description	The AI module analyzes each student's financial history and academic records to determine the level of risk associated with granting the loan. This assessment assists the system administrator in making the final decision on loan approval.

2.2 Fully-Dressed Use Cases

2.2.1 Use Case: Apply for Loan

Use Case	Apply for Loan
Scope	ScholarChain
Level	User Level
Primary Actor(s):	Student
Stakeholders and Interests	Student: The Student wants to apply for an interest-free loan.
Preconditions	1. The student must be registered on the ScholarChain platform and must be logged in to start the application process. 2. The students must have all required documents prepared and ready for uploading.
Success Guarantee	1. Upon successful completion, the student's loan application is submitted. 2. The student receives confirmation of submission.

	Actor Action	System Response
Main Success Scenario	Student logs into the Scholar-Chain platform.	System authenticates the student and provides access to their dashboard.
	Student selects the "Apply for Loan" option from the dashboard.	System displays the loan application form with sections for personal, academic, and financial details.
	Student fills in their personal information, academic details, and financial status.	System verifies the entered details for completeness.
	Student uploads the required documents.	System checks that the documents are in the correct format (e.g., PDF) and stores them securely.
	Student reviews the form and submits the application.	System saves the application and sends a confirmation notification to the student.
Alternative Scenarios	1. Invalid Login 2. Incomplete Form 3. Missing Documents	

2.2.2 Use Case: Provide Funding

Use Case	Provide Funding
Scope	ScholarChain
Level	User Level
Primary Actor(s):	Social Welfare Entity (Donor)
Stakeholders and Interests	Donor: Wants an easy-to-use and transparent to contribute to the education of students.

Preconditions	1. The donor must be registered and logged in to the ScholarChain platform.	
Success Guarantee	1. Upon successful transaction, the donation is added to the immutable blockchain ledger.	
Main Success Scenario	**Actor Action**	**System Response**
	Donor logs into the ScholarChain platform.	System authenticates the donor and provides access to their dashboard.
	Student selects the "Contribute Funds" option from the dashboard.	The system displays funding options
	Donor enters the desired contribution amount.	
	Donor provides payment details and confirms the contribution.	The system processes the payment securely and confirms the transaction.
	Donor receives confirmation of successful payment.	The system updates the funding pool, records the transaction in the blockchain ledger, and displays the current status of contributions.
Alternative Scenarios	1. Payment Failure 2. Incorrect Contribution Amount	

2.2.3 Use Case: Approve Loan

Use Case	Approve Loan
Scope	ScholarChain
Level	User Level

<table>
<tr><td>Primary Actor(s):</td><td>System Administrator</td></tr>
<tr><td>Stakeholders and Interests</td><td>Student: Wants their loan application to be approved as soon as possible.
System Administrator: Needs to ensure that all documentation is in order and whether the application is deserving of loan before giving approval.</td></tr>
<tr><td>Preconditions</td><td>1. All required documents must be successfully uploaded and verified.</td></tr>
<tr><td>Success Guarantee</td><td>1. The loan application is approved, and the student is notified of the disbursement schedule.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>The AI does risk assessment on the student's application.</td><td>The system stores the results of the risk assessment to be reviewed by the system administrator.</td></tr>
<tr><td>The system administrator reviews the risk assessment results.</td><td></td></tr>
<tr><td>The system administrator approves the loan application.</td><td>The system updates the loan status to "Approved" and notifies the student.</td></tr>
</table>
</td></tr>
<tr><td>Alternative Scenarios</td><td>1. Missing Verification</td></tr>
</table>

2.2.4 Use Case: Track Repayment

Use Case	Track Repayment
Scope	ScholarChain
Level	User Level
Primary Actor(s):	Student, Parent/Guardian

Stakeholders and Interests	Student: Wants to ensure timely repayments and track the repayment schedule in order to manage finances accordingly.
Preconditions	1. The student or parent/guardian must be logged in to ScholarChain. 2. The loan must be approved and disbursed in order to track repayments.
Success Guarantee	1. The students are able to track their repayment schedule and receive reminders about upcoming repayments.
Main Success Scenario	**Actor Action** / **System Response** Student logs into ScholarChain. / The system authenticates the user and displays their dashboard. User navigates to the "Repayment Tracking" section. / The system displays detailed repayment information, including due dates, amounts, and history.
Alternative Scenarios	1. Missed payment 2. Partial payment 3. Payment processing failure

2.2.5 Use Case: Customer Support Interaction

Use Case	Customer Support Interaction
Scope	ScholarChain
Level	User Level
Primary Actor(s):	Student, System Administrator

<table>
<tr><td>Stakeholders and Interests</td><td>Student: Wants to be able to receive help with any problems related to the platform, and be able to request human support in case of a complex problem for AI.
System Administrator: Needs to assist students with issues the AI cannot resolve.</td></tr>
<tr><td>Preconditions</td><td>1. The student must be logged in to the ScholarChain platform to recieve support.</td></tr>
<tr><td>Success Guarantee</td><td>1. The student's question or issue is resolved, either by the AI assistant or with manual intervention by a system administrator.</td></tr>
<tr><td>Main Success Scenario</td><td><table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Student requests support.</td><td>The system displays available options, including an AI-powered chatbot.</td></tr>
<tr><td>Student asks a question via the chatbot.</td><td>The chatbot analyzes the question and provides a response based on available information.</td></tr>
<tr><td>Student is unsatisfied with the response or the issue is complex.</td><td>The system escalates the request to a system administrator for manual assistance.</td></tr>
<tr><td>System administrator reviews the request and responds.</td><td>The system notifies the student and displays the response from the administrator.</td></tr>
</table></td></tr>
<tr><td>Alternative Scenarios</td><td>1. No Available Administrator
2. AI inference error</td></tr>
</table>

2.2.6 Use Case: Personalized Repayment Plan Generation

<table>
<tr><td>Use Case</td><td>Personalized Repayment Plan Generation</td></tr>
<tr><td>Scope</td><td>ScholarChain</td></tr>
<tr><td>Level</td><td>User Level</td></tr>
<tr><td>Primary Actor(s):</td><td>Student</td></tr>
<tr><td>Stakeholders and Interests</td><td>Student: Wants a flexible repayment schedule based on their current financial needs.</td></tr>
<tr><td>Preconditions</td><td>1. The student must have an approved loan.
2. The student must have submitted all relevant documentation and details</td></tr>
<tr><td>Success Guarantee</td><td>1. The system generates a personalized repayment plan, and the student is notified with the details.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td></td><td>AI generates a repayment plan with due dates and amounts that are affordable for the student.</td></tr>
<tr><td>Student reviews and accepts the repayment plan.</td><td>The system records the acceptance and sets up reminders for payment schedules.</td></tr>
<tr><td>Student is unsatisfied with the response or the issue is complex.</td><td>The system escalates the request to a system administrator for manual assistance.</td></tr>
</table>
</td></tr>
<tr><td>Alternative Scenarios</td><td>1. Outdated or missing documents</td></tr>
</table>

2.2.7 Use Case: Automated Loan Disbursement

Use Case	Automated Loan Disbursement
Scope	ScholarChain

<table>
<tr><td>Level</td><td>User Level</td></tr>
<tr><td>Primary Actor(s):</td><td>System Administrator, Blockchain Ledger</td></tr>
<tr><td>Stakeholders and Interests</td><td>Student: Wants to receive loans as soon as possible after loan approval.</td></tr>
<tr><td>Preconditions</td><td>1. The loan must be fully approved, and the student's agreement must be signed.
2. All requirements for smart contracts must be met.</td></tr>
<tr><td>Success Guarantee</td><td>1. Loan funds are successfully disbursed to the student's designated account.</td></tr>
<tr><td>Main Success Scenario</td><td><table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>The smart contract validates the loan approval conditions.</td><td>The system verifies all criteria are met for disbursement.</td></tr>
<tr><td>Upon successful validation, funds are transferred to the student's account.</td><td>The system updates the blockchain ledger with the transaction details.</td></tr>
<tr><td>Student is notified of the disbursement.</td><td>The system sends a notification with details on the amount disbursed and the repayment schedule.</td></tr>
</table></td></tr>
<tr><td>Alternative Scenarios</td><td>1. Smart Contract Failure</td></tr>
</table>

2.2.8 Use Case: Verify Loan Agreement

Use Case	Verify Loan Agreement
Scope	ScholarChain
Level	User Level
Primary Actor(s):	Student, Parent/Guardian

Stakeholders and Interests	Student: Wants to be aware of the terms and conditions of the loan agreement before signing it. Parent/Guardian: Wants to ensure that the loan agreement is fair and that all obligations are clearly stated.
Preconditions	1. The loan must be approved and ready for signing.
Success Guarantee	1. The loan agreement is reviewed and signed digitally by the student.
Main Success Scenario	<table><tr><th>Actor Action</th><th>System Response</th></tr><tr><td>The student receives a notification that the loan agreement is ready for review.</td><td>The system displays the loan agreement document.</td></tr><tr><td>Student reviews the loan terms and conditions.</td><td>The system provides options to accept or request clarification.</td></tr><tr><td>student digitally signs the loan agreement.</td><td>The system records the signatures and marks the agreement as complete.</td></tr></table>
Alternative Scenarios	1. Request for Changes 2. Signature Declined

2.2.9 Use Case: Payment Scheduling and Notification

Use Case	Payment Scheduling and Notification
Scope	ScholarChain
Level	User Level
Primary Actor(s):	Student
Stakeholders and Interests	Student: wants to be aware of all upcoming payments.

<table>
<tr><td>Preconditions</td><td colspan="2">1. The repayment plan must already be generated and accepted by the student.</td></tr>
<tr><td>Success Guarantee</td><td colspan="2">1. Students receive notifications regarding upcoming payments.</td></tr>
<tr><td rowspan="3">Main Success Scenario</td><td>Actor Action</td><td>System Response</td></tr>
<tr><td>Student receives an alert about an upcoming payment.</td><td>The system provides details on the payment due date, amount, and instructions on how to make the payment.</td></tr>
<tr><td>Student decides to make the payment before the due date.</td><td>The system processes the payment and updates the repayment history accordingly.</td></tr>
<tr><td>Alternative Scenarios</td><td colspan="2">1. Missed Notification
2. Payment Postponement Request</td></tr>
</table>

2.3 Use Case Diagram

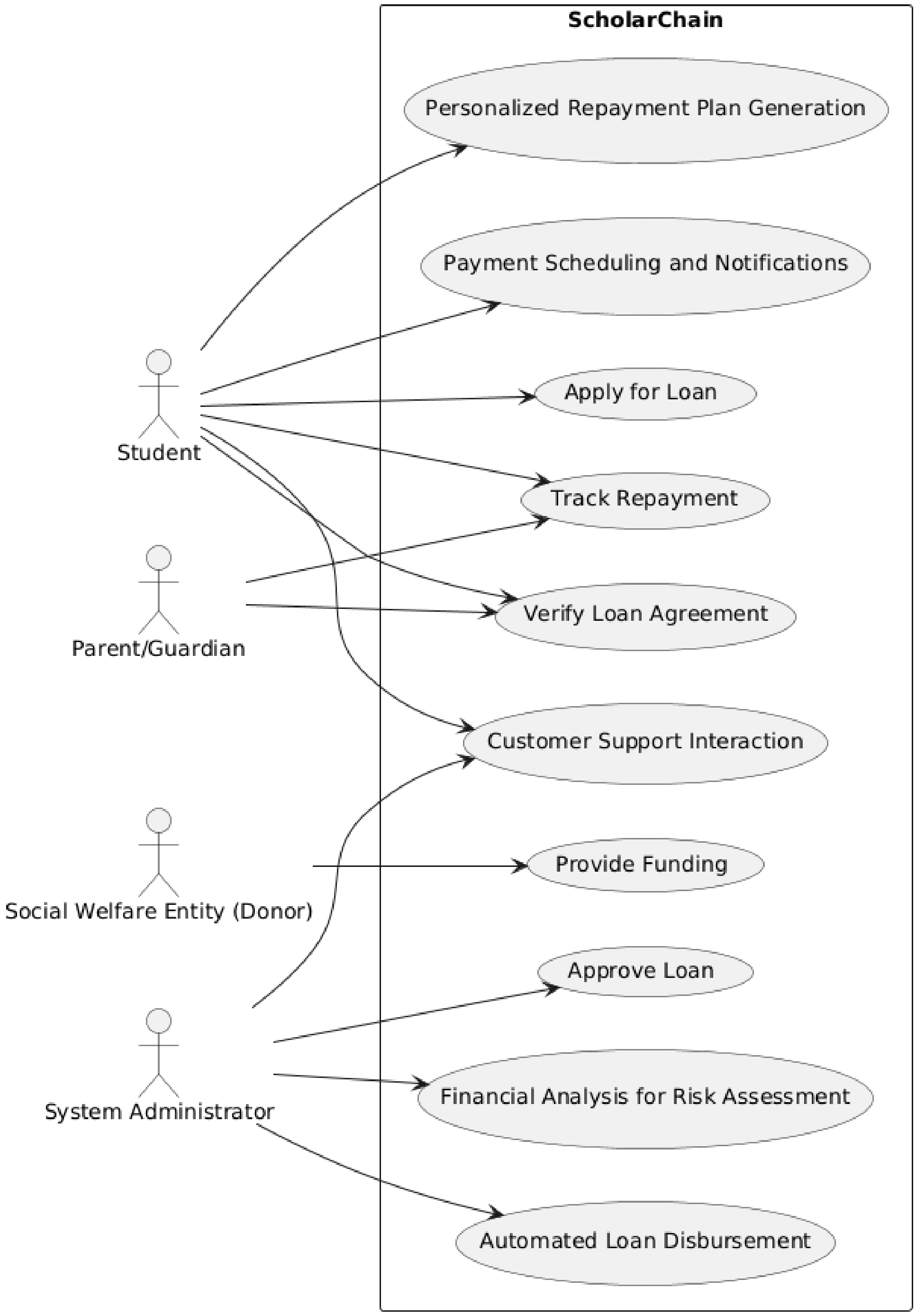

Figure 2.1: ScholarChain Use Case Diagram

2.4 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style.

2.4.1 Module 1: Student Interaction Module

The following are the requirements for Module 1:

1. **Loan Application Submission:** Students can submit loan applications by providing personal, acedemic, and financial information.
2. **Personalized Repayment Plan Schedule:** System assesses the current financial situation of the user and generates a personalized repayment plan.
3. **Repayment Tracking:** Students can track their upcoming payments and view payment history.
4. **AI-Powered Customer Support:** Students can interact with an AI-powered chatbot in order to address their queries regarding their application and the loan process.

2.4.2 Module 2: Funding and Loan Management Module

Following are the requirements for module 2:

1. **Funds Provision by Donors:** Donors can contribute funds to the platform via secure payment gateways.
2. **Loan Application Approval:** The system administrator reviews the AI-assessed application and makes the final decision on whether to approve or decline.
3. **Automated Loan Disbursement:** Funds are automatically disbursed via smart contracts one applications are approved.

2.4.3 Module 3: Administrative Tools

Following are the requirements for module 3:

1. **Automated Risk Assessment:** AI evaluates each application and assigns them a loan risk score.

2. **Verification of Loan Signature:** System administrators verify that an application's terms and conditions have been accepted by the student and has digitally signed the agreement.

2.4.4 Module 4: Notifications and Reminders

1. **Payment Scheduling and Alerts:** Automatic notifications are sent to students to remind them of upcoming payment deadlines.
2. **Loan Status Update:** The system notifies students about the current status of their loan application.

2.5 Non-Functional Requirements

This section outlines the non-functional requirements for the ScholarChain platform.

2.5.1 Reliability

ScholarChain must be highly reliable. The target mean time between failures should be no less than 500 hours. This will ensure continuous availability for the students and donors who wish to use the platform. In case of failure, automatic error detection mechanisms should be in place, as well as data protection measures to ensure that the user's data is secure. A detailed failure log should be maintained for debugging and future improvement.

- **Failure Tolerance:** The system should not experience more than one failure per month. The recovery period should not be more than 1 hour.

2.5.2 Usability

The ScholarChain platform must be easy to use for all types of users (Students, Parents, Donors). The interface must be intuitive and easy to navigate without much hassle.

- **Ease of Learning:** A new user should be able to easily understand and navigate the platform within a few minutes of interaction.
- **Error Prevention and Recovery:** Common errors such as incomplete form and uploading files in incorrect format should be clearly highlighted.
- **Interaction Efficiency:** All tasks such as submitting an application should be completable withing 5 steps.

2.5.3 Performance

The ScholarChain system should perform efficiently to provide a smooth user experience under different load conditions.

- **Response Time:** 90% of all pages must be loaded completely within 3 seconds over a 20Mbps connection.
- **Scalability:** The platform must be able to support almost 5000 concurrent users without degradation in performance.
- **Data Processing:** The data for the application process should be processable by the AI within 30 seconds of submission.

2.5.4 Security

The ScholarChain platform must be highly secure to protect user data and maintain the integrity of the system.

- **Data Protection:** All user data must be encrypted at all times.
- **Access Control:** Different users must have different access rights ensuring unauthorized users cannot access sensitive information and functionalities.

2.6 Domain Model

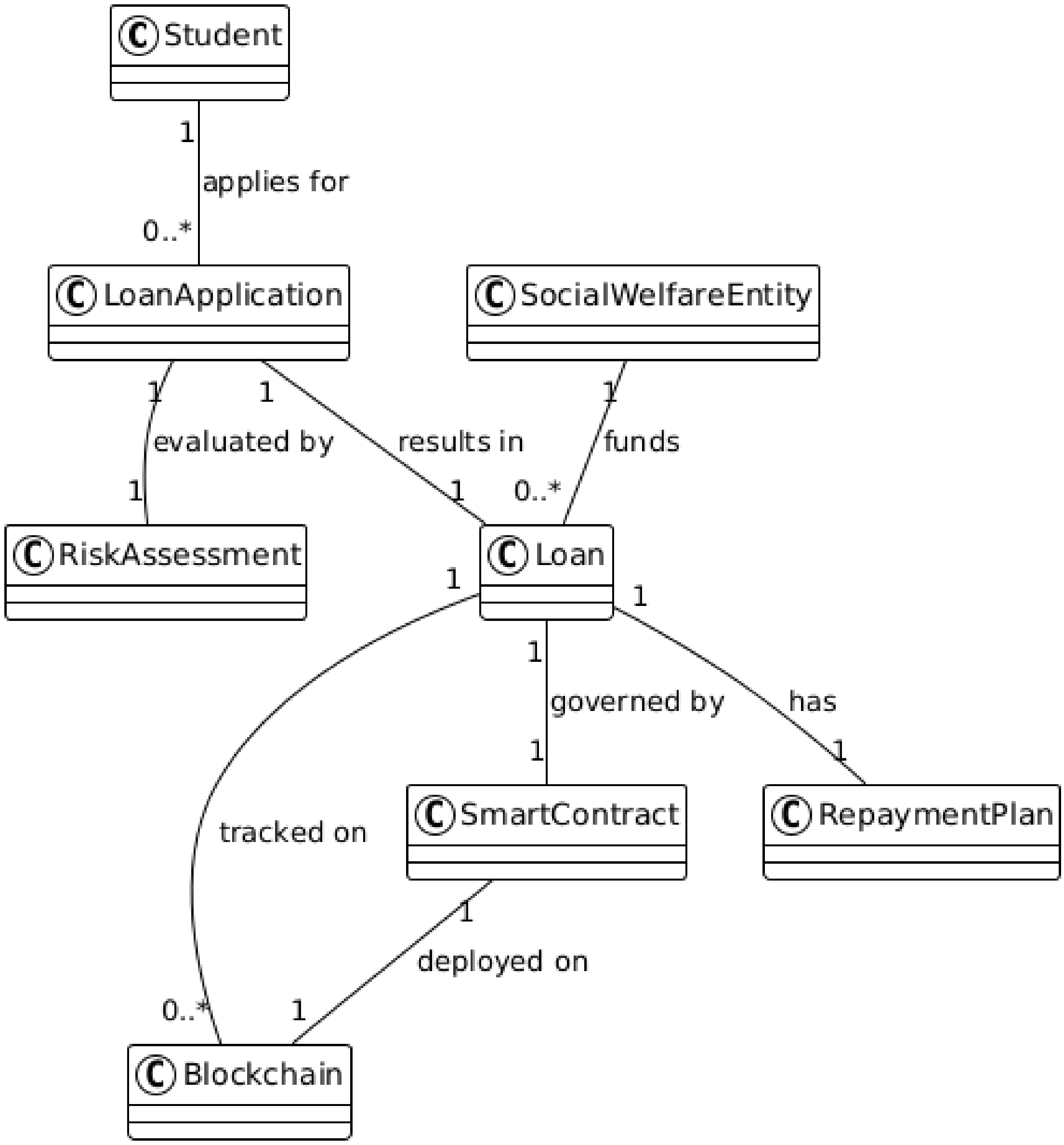

Figure 2.2: ScholarChain Domain Model

Chapter 3

System Overview

3.1 Architectural Design

3.1.0.1 Box and Line Diagram

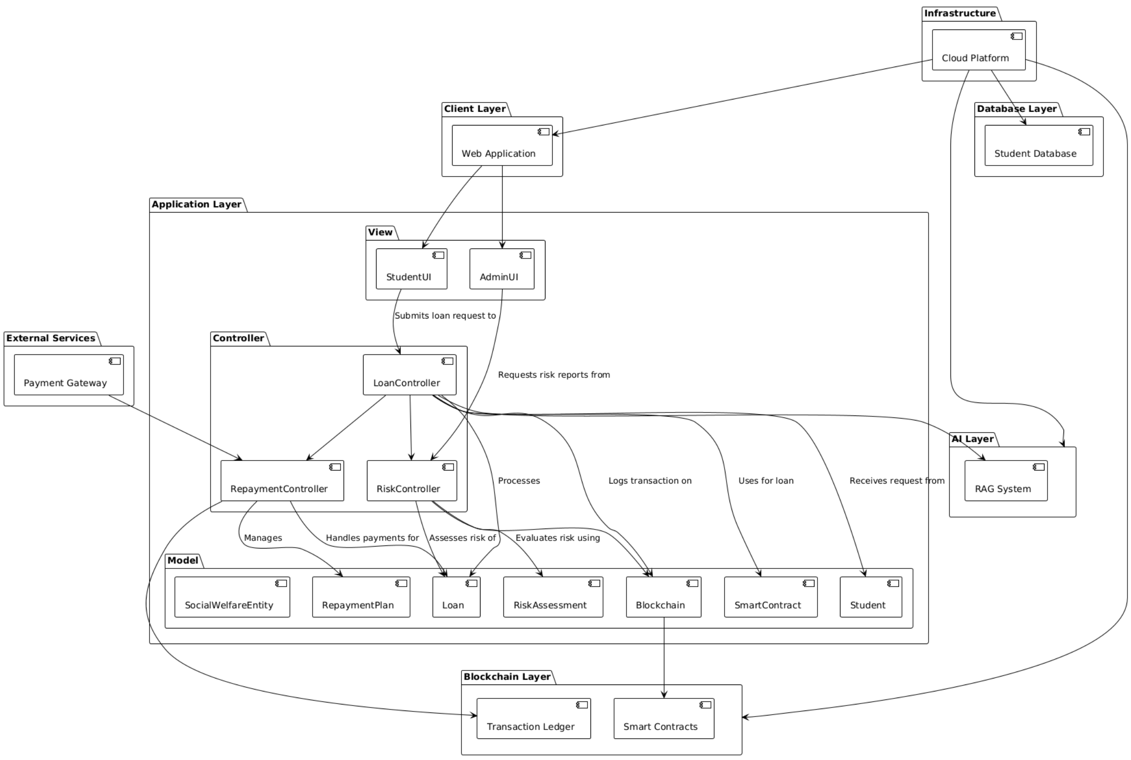

Figure 3.1: Box and Line Diagram

The box-and-line diagram is a high-level architectural representation of ScholarChain, showing how system layers interact, such as the **Client Layer**, **Application Layer**, **Database Layer**, **AI layer**, and **Blockchain Layer**.

3.1.1 Component Diagram

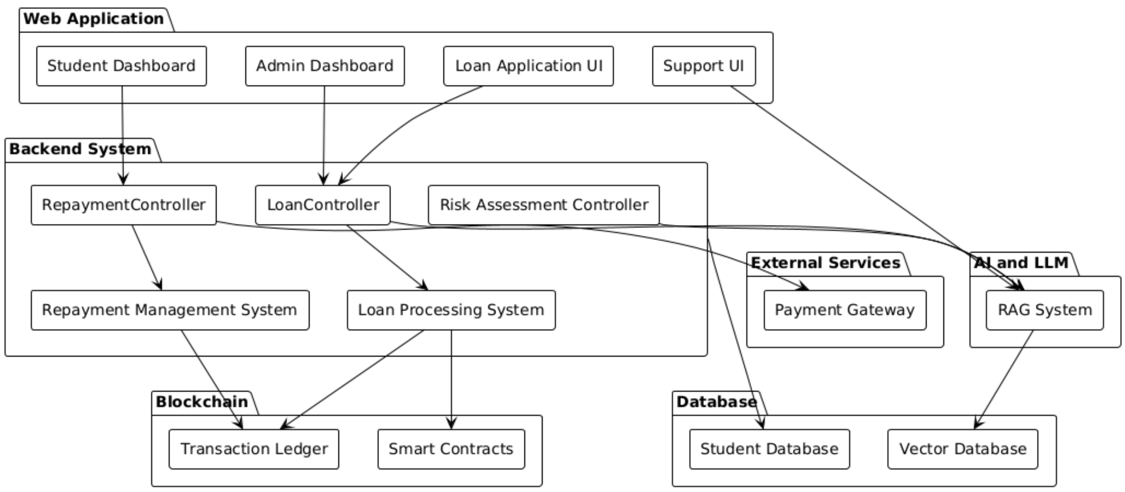

Figure 3.2: component Diagram

The Component Diagram shows a detailed architecture of ScholarChain. This diagram provides a detailed understanding of how each subsystem contributes to the functionality of the system. Connections between the different components and layers are highlighted in this diagram.

3.2 Design Models

3.2.1 Activity Diagram

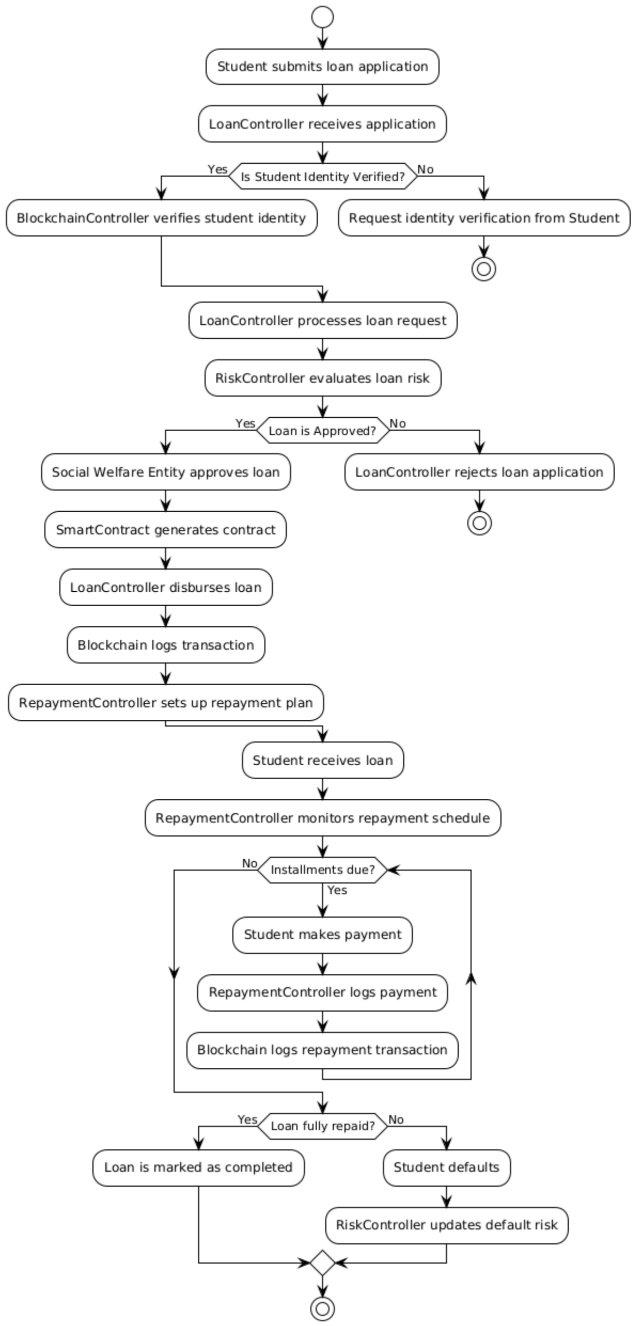

Figure 3.3: activity Diagram

The Activity Diagram details the workflow of a student's journey through the Scholar-Chain platform.

3.2.2 Class Diagram

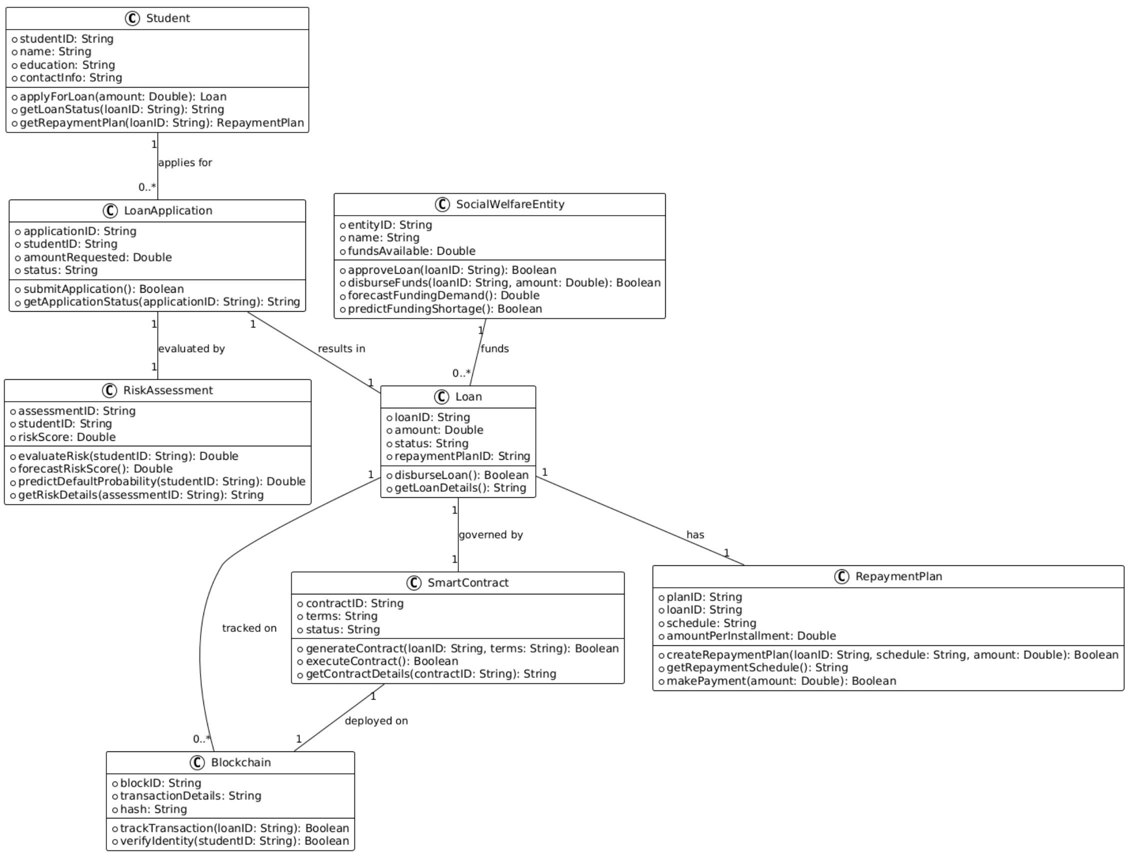

Figure 3.4: Class Diagram

The Class Diagram shows how the system's components are modeled as classes in an object-oriented design.

3.2.3 Sequence Diagrams

3.2.3.1 Loan Application Sequence Diagram

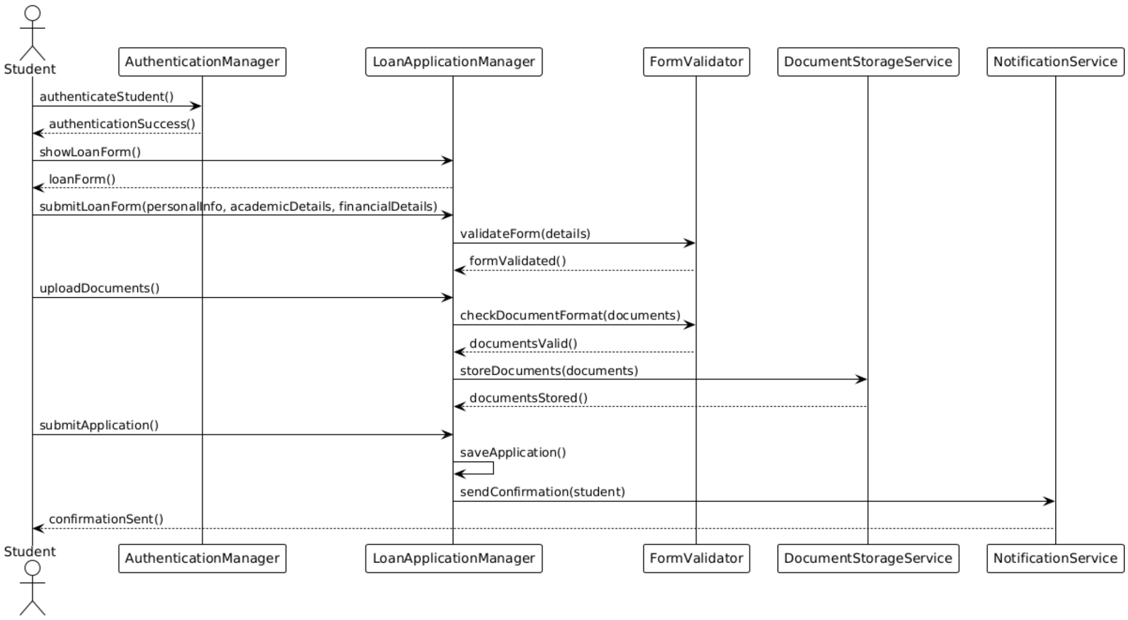

Figure 3.5: Loan Application Sequence Diagram

3.2.3.2 Funding Contribution Sequence Diagram

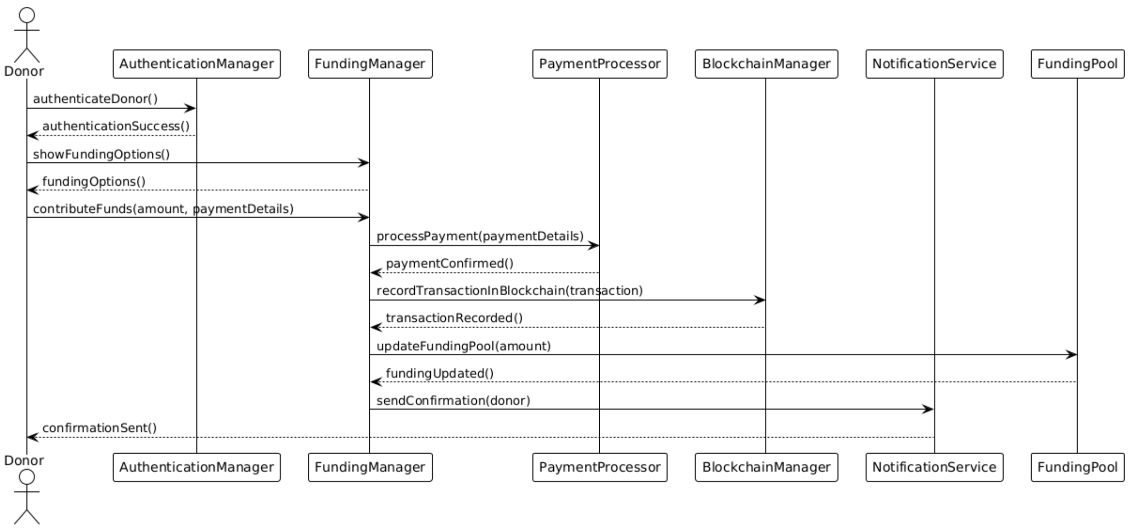

Figure 3.6: Funding Contribution Sequence Diagram

3.2.3.3 Approve Loan Sequence Diagram

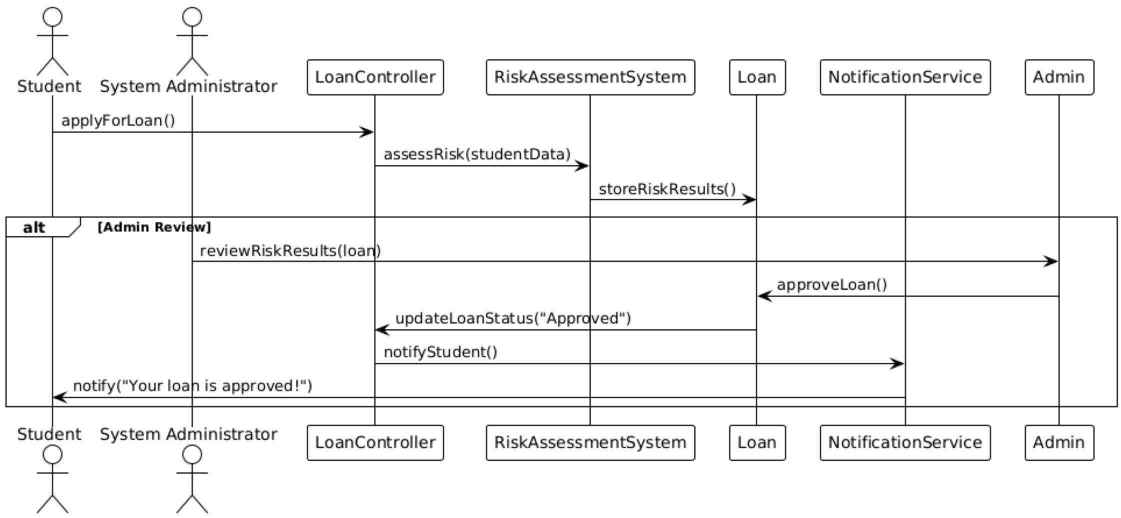

Figure 3.7: Approve Loan Sequence Diagram

3.2.3.4 Track Repayment Sequence Diagram

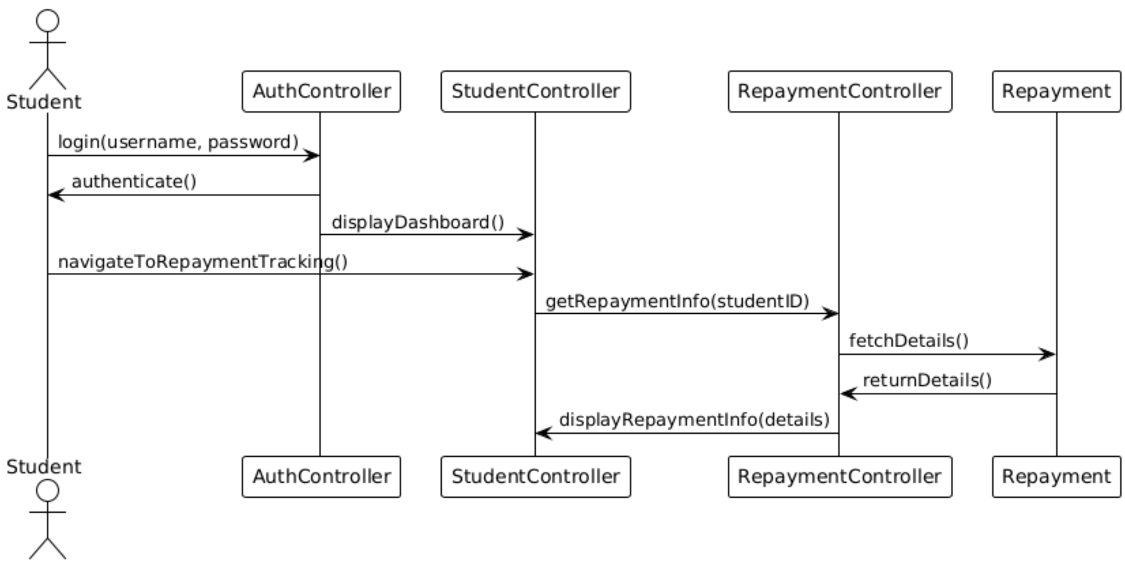

Figure 3.8: Track Repayment Sequence Diagram

3.2.3.5 Customer Support Sequence Diagram

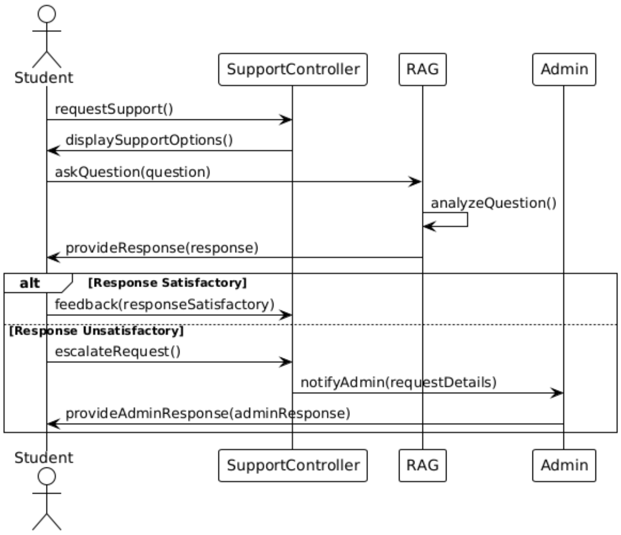

Figure 3.9: Customer Support Sequence Diagram

3.2.3.6 Payment Scheduling and Notification Sequence Diagram

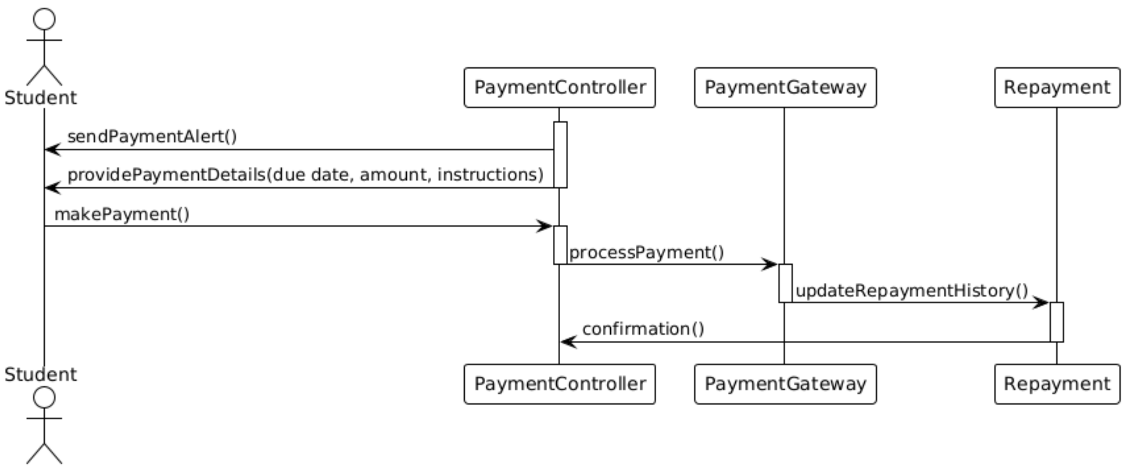

Figure 3.10: Payment Scheduling and Notification Sequence Diagram

3.2.3.7 Personalized Repayment Sequence Diagram

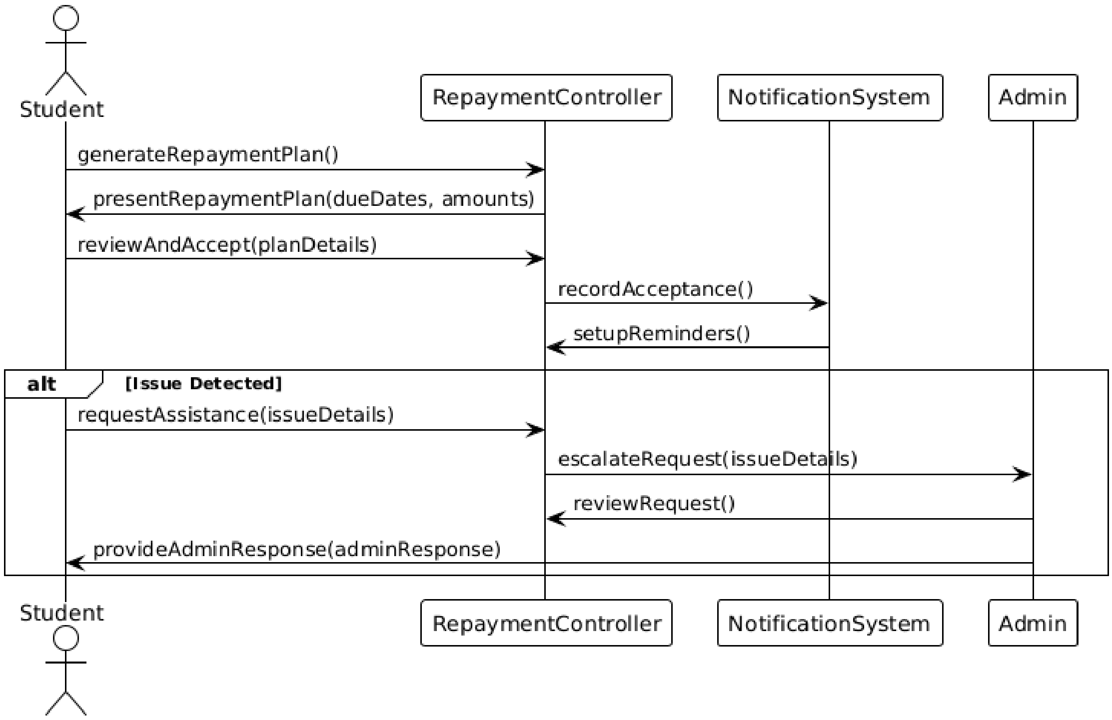

Figure 3.11: Personalized Repayment Sequence Diagram

3.2.3.8 Automated Loan Disbursement Sequence Diagram

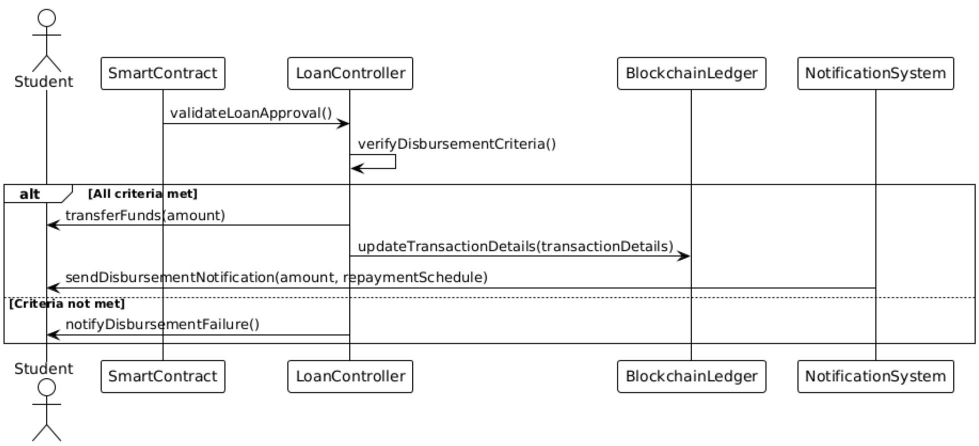

Figure 3.12: Automated Loan Disbursement Sequence Diagram

3.2.3.9 Verify Loan Sequence Diagram

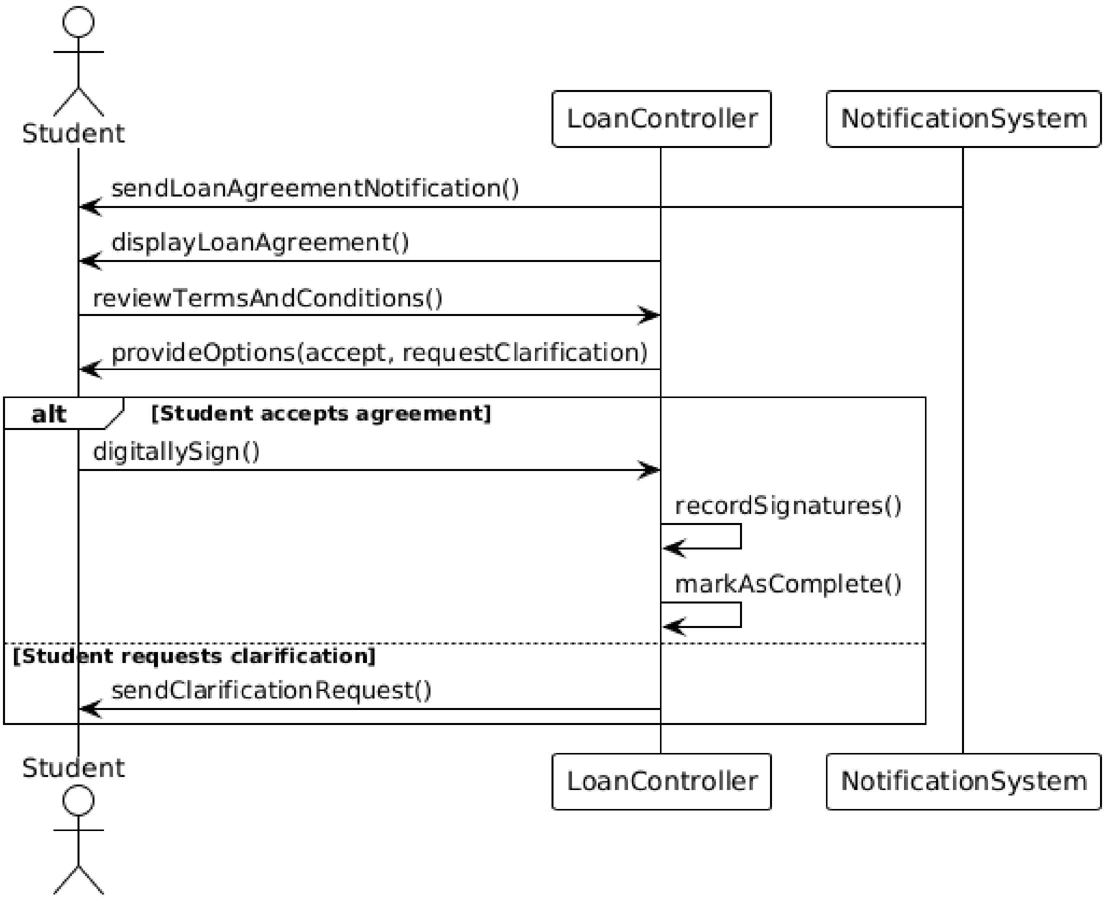

Figure 3.13: Verify Loan Sequence Diagram

3.2.4 State Transition Diagram

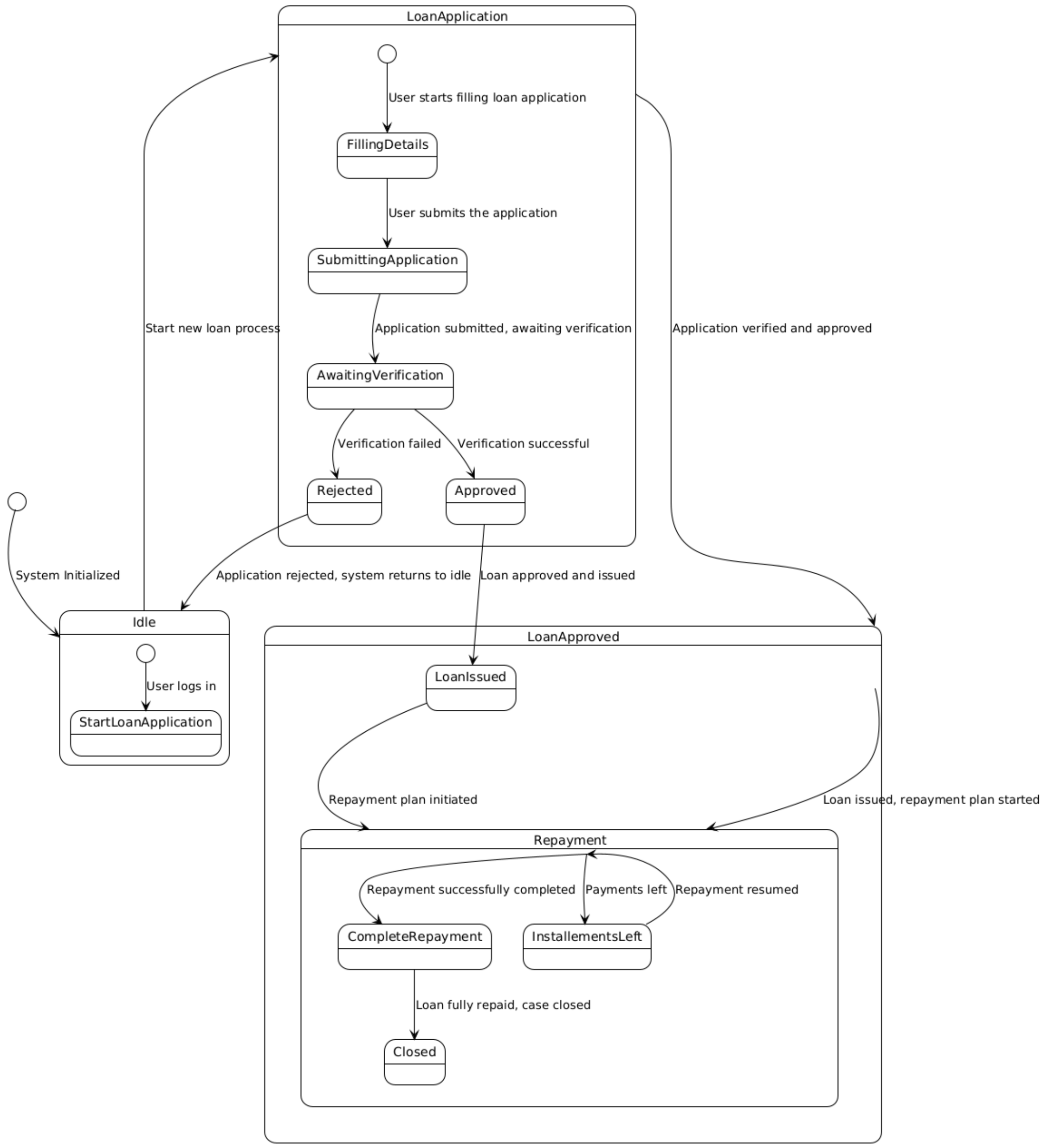

Figure 3.14: State Transition Diagram

The State Transition Diagram describes how an application progresses through different states.

3.3 Data Design

3.3.1 Schema Diagram

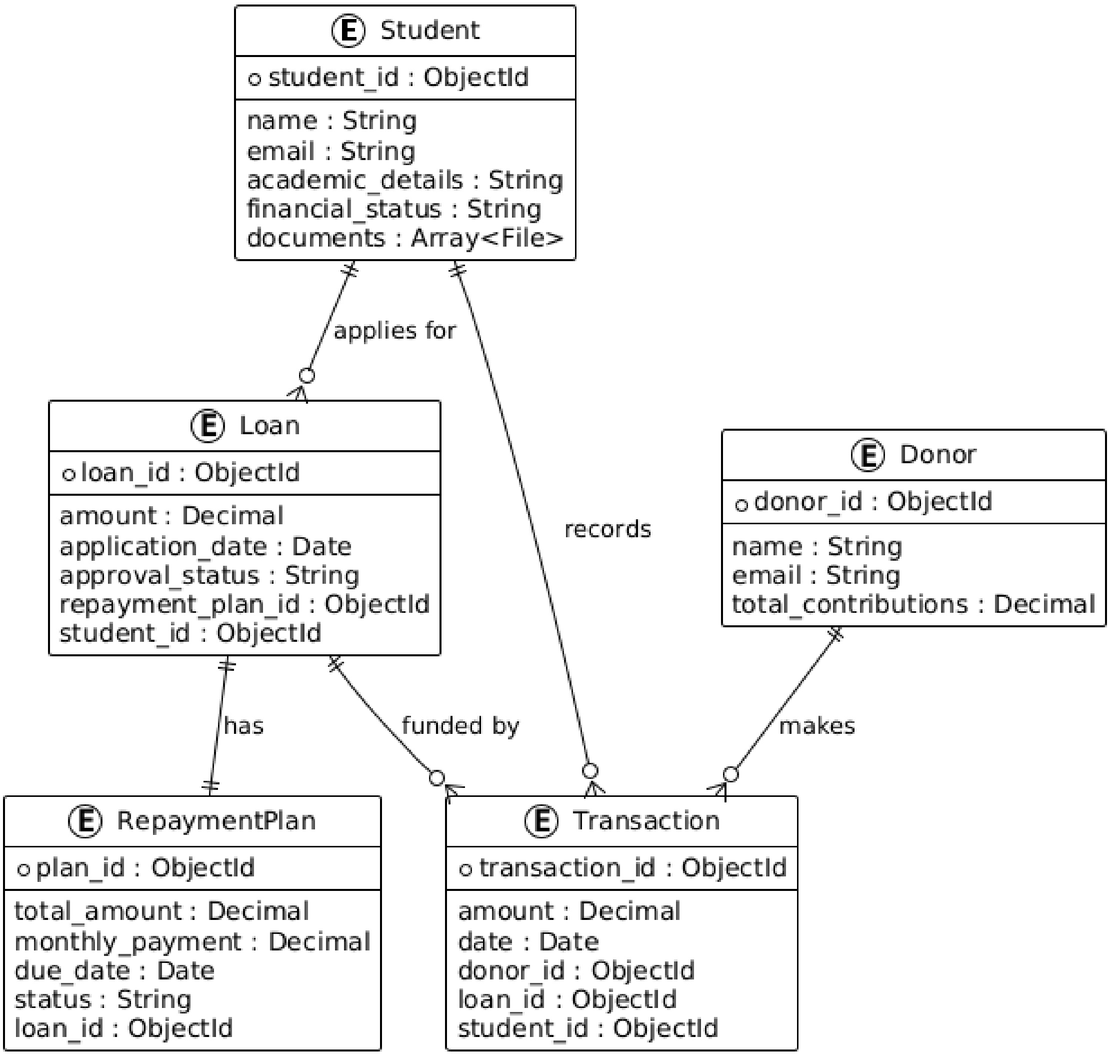

Figure 3.15: Schema Diagram

The Schema Diagram provides a detailed look at how different entities within ScholarChain are interrelated.

3.3.2 Json Schema

Student Object

```
{
  "bsonType": "object",
  "required": ["student_id", "name", "email", "
     academic_details", "financial_status"],
  "properties": {
    "student_id": {
      "bsonType": "objectId"
    },
    "name": {
      "bsonType": "string"
    },
    "email": {
      "bsonType": "string"
    },
    "academic_details": {
      "bsonType": "string"
    },
    "financial_status": {
      "bsonType": "string"
    },
    "documents": {
      "bsonType": "array",
      "items": {
        "bsonType": "string"
      }
    }
  }
}
```

Loan Schema:

```
{
  "bsonType": "object",
  "required": ["loan_id", "amount", "application_date", "
     approval_status", "repayment_plan_id", "student_id"
     ],
  "properties": {
    "loan_id": {
```

```
        "bsonType": "objectId"
      },
      "amount": {
        "bsonType": "decimal"
      },
      "application_date": {
        "bsonType": "date"
      },
      "approval_status": {
        "bsonType": "string"
      },
      "repayment_plan_id": {
        "bsonType": "objectId"
      },
      "student_id": {
        "bsonType": "objectId"
      }
    }
}
```

RepaymentPlan Schema:

```
{
  "bsonType": "object",
  "required": ["plan_id", "total_amount", "
     monthly_payment", "due_date", "status", "loan_id"],
  "properties": {
    "plan_id": {
      "bsonType": "objectId"
    },
    "total_amount": {
      "bsonType": "decimal"
    },
    "monthly_payment": {
      "bsonType": "decimal"
    },
    "due_date": {
      "bsonType": "date"
    },
    "status": {
      "bsonType": "string"
```

```
    },
    "loan_id": {
      "bsonType": "objectId"
    }
  }
}
```

Donor Schema:

```
{
  "bsonType": "object",
  "required": ["donor_id", "name", "email", "
     total_contributions"],
  "properties": {
    "donor_id": {
      "bsonType": "objectId"
    },
    "name": {
      "bsonType": "string"
    },
    "email": {
      "bsonType": "string"
    },
    "total_contributions": {
      "bsonType": "decimal"
    }
  }
}
```

Transaction Schema:

```
{
  "bsonType": "object",
  "required": ["transaction_id", "amount", "date", "
     donor_id", "loan_id", "student_id"],
  "properties": {
    "transaction_id": {
      "bsonType": "objectId"
    },
    "amount": {
      "bsonType": "decimal"
    },
```

```
    "date": {
      "bsonType": "date"
    },
    "donor_id": {
      "bsonType": "objectId"
    },
    "loan_id": {
      "bsonType": "objectId"
    },
    "student_id": {
      "bsonType": "objectId"
    }
  }
}
```

Bibliography

[1] Gaper.io. Automated underwriting systems: How llms can automate loan processing and increase approval rates, May 2024. Accessed: 2024-08-20.